Use Case Flow Document

Web-Based ML Model Evaluator

1. Introduction

This document details the flow of events for the use cases shown in the Use Case Diagram. Each flow records its goal, actor, main flow, alternate/exception flows, and postconditions, and references the SRS requirement IDs.

2. Actors

Actor	Description
Student / Registered User	Performs the primary dataset-to-model workflow on resources they are authorized to access.
Instructor	Views only evaluation results explicitly made available for review.
Admin	Performs the administrative user, dataset-record, and activity-log functions defined in the SRS.
ML Service	Executes model-training work when invoked by the application.

3. Use Case Summary

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ID	Use Case	Primary Actor(s)	Related Requirement(s)
UC-01	Register / Login / Logout	Student / Registered User	FR-1.1–FR-1.4
UC-02	Upload Dataset	Student / Registered User	FR-2.1, FR-2.2, FR-2.5
UC-03	Preview Dataset and Statistics	Student / Registered User	FR-2.3, FR-2.4
UC-04	Preprocess Data	Student / Registered User	FR-3.1
UC-05	Select Target Column	Student / Registered User	FR-3.2
UC-06	Set Train-Test Split	Student / Registered User	FR-3.3
UC-07	Select ML Algorithm	Student / Registered User	FR-4.1
UC-08	Train Model	Student / ML Service	FR-4.2, FR-4.3
UC-09	Evaluate Results and View Run History	Student; Instructor (read-only)	FR-5.1, FR-5.2, FR-6.1, FR-7.4
UC-10	Compare Models	Student / Registered User	FR-5.3
UC-11	Download Model / Report	Student / Registered User	FR-5.4
UC-12	Manage Users	Admin	FR-7.1
UC-13	Manage Dataset Records	Admin	FR-7.2
UC-14	View System Logs	Admin	FR-7.3

4. Use Case Relationships

- UC-01 is a precondition for protected Student and Instructor actions; Admin actions also require authenticated Admin access.
- UC-08 uses the selected dataset, preprocessing configuration, target column, train-test split, and selected algorithm established in UC-02 and UC-04 to UC-07.
- UC-09 follows a completed training run (UC-08); an Instructor may view only results explicitly made available for review.
- UC-10 and UC-11 are optional actions after results are available; they are not mandatory steps in every evaluation.

5. Detailed Use Case Flows

UC-01 Register / Login / Logout

Field	Description
Goal	Authenticate to use protected functions.
Actor	Student / Registered User
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Open the application. 2. Register or enter credentials. 3. System validates input and credentials. 4. Valid login opens the user's workspace. 5. Logout terminates the session.
Alternate Flow	Missing registration fields are rejected; invalid credentials do not create a session; a logged-out session cannot access protected resources.

UC-02 Upload Dataset

Field	Description
Goal	Upload a CSV for evaluation.
Actor	Student / Registered User
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Open upload. 2. Select a CSV. 3. System validates file type, readability, structure, and required data. 4. System associates metadata with the owner. 5. System confirms upload.
Alternate Flow	Unsupported, empty, malformed, unreadable, or over-limit input is rejected with feedback; existing data is not corrupted.

UC-03 Preview Dataset and Statistics

Field	Description
Goal	Inspect the dataset before configuring an experiment.
Actor	Student / Registered User
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Select an owned dataset. 2. System displays column names and a bounded row sample. 3. System displays row/column counts and available type/missing-value information.
Alternate Flow	Unknown or unauthorized dataset identifiers are denied; parsing failures are reported without exposing internal details.

UC-04 Preprocess Data

Field	Description
Goal	Prepare data using supported options.
Actor	Student / Registered User
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Select a dataset. 2. Choose supported preprocessing options. 3. System validates the configuration. 4. System applies or records the valid configuration.
Alternate Flow	Unsupported or contradictory options are rejected; a preprocessing failure is reported and does not corrupt prior runs.

UC-05 Select Target Column

Field	Description
Goal	Choose the prediction target.
Actor	Student / Registered User
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Open the target selector. 2. System lists dataset columns. 3. Select a column. 4. System validates that the column exists and is usable.
Alternate Flow	Missing, nonexistent, or unusable target selection is rejected with a clear message.

UC-06 Set Train-Test Split

Field	Description
Goal	Configure the training and test partitions.
Actor	Student / Registered User
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Open split configuration. 2. Select a supported proportion. 3. System validates the value and partition sizes. 4. Save configuration.
Alternate Flow	Unsupported proportions or unusable partition sizes are rejected.

UC-07 Select ML Algorithm

Field	Description
Goal	Choose a compatible supported algorithm.
Actor	Student / Registered User
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Open algorithm selection. 2. System lists supported algorithms. 3. Select an algorithm. 4. System validates task/data compatibility.
Alternate Flow	Unsupported or incompatible algorithms are rejected by the backend.

UC-08 Train Model

Field	Description
Goal	Train using the configured experiment.
Actor	Student / ML Service

Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Start training. 2. Backend validates access and configuration. 3. Backend invokes the training component. 4. System shows status. 5. System records completion/failure and associates results with the correct user.
Alternate Flow	Invalid configuration prevents training; a training failure is reported and does not alter previous completed runs.

UC-09 Evaluate Results and View Run History

Field	Description
Goal	Review metrics, visualizations, and saved runs.
Actor	Student / Registered User; Instructor (read-only)
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Open run history or a completed run. 2. System checks authorization. 3. Display status, configuration, applicable metrics, and supported visualizations. 4. Instructor views only results made available for review.
Alternate Flow	Unauthorized run access is denied; unavailable metrics are not presented as valid; Instructor write attempts are denied.

UC-10 Compare Models

Field	Description
Goal	Compare two or more available model runs.
Actor	Student / Registered User
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Select at least two accessible runs. 2. System checks comparability. 3. Display comparable metrics and configuration details.
Alternate Flow	Fewer than two runs or incompatible tasks are handled with clear feedback; unauthorized runs are excluded/denied.

UC-11 Download Model / Report

Field	Description
Goal	Download a supported artifact or evaluation report.
Actor	Student / Registered User
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Select a completed run. 2. Choose a supported download. 3. System checks ownership and availability. 4. Return the selected artifact/report.
Alternate Flow	Unsupported format, missing artifact, export failure, or unauthorized run results in a clear error/denial.

UC-12 Manage Users

Field	Description
Goal	Admin views accounts and performs supported account actions.
Actor	Admin
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Admin authenticates. 2. Opens user management. 3. Reviews available user records. 4. Performs a supported action. 5. System validates and confirms it.
Alternate Flow	Non-Admin access is denied; invalid account actions are rejected.

UC-13 Manage Dataset Records

Field	Description
Goal	Admin views/manages stored dataset records.
Actor	Admin
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Admin authenticates. 2. Opens dataset management. 3. Reviews metadata. 4. Performs a supported action. 5. System preserves owner and run associations.
Alternate Flow	Non-Admin access is denied; operations that break ownership or run references are rejected.

UC-14 View System Logs

Field	Description
Goal	Admin reviews activity logs in read-only form.
Actor	Admin
Preconditions	Relevant account, dataset, run, or configuration exists; the actor is authenticated where required and has the required permission.
Main Flow	1. Admin authenticates. 2. Opens logs. 3. System displays available activity

	entries. 4. Admin reviews them.
Alternate Flow	Non-Admin access is denied; unavailable logs produce a controlled error; secrets and dataset contents are not exposed.